

AUDIO-TEXT JAILBREAK ATTACK ON LARGE AUDIO-LANGUAGE MODELS: TOWARDS GENERALITY AND STEALTHINESS

Yuhong Li^{1,2}, Jiabao Zhang^{1,2}, Yan Chen², Zhihui Zhao^{1,2}, Jianhua Wang^{1,2,*}

¹Shanxi Key Laboratory of Industrial Internet Security, Taiyuan University of Technology, China

²College of Computer Science and Technology, Taiyuan University of Technology, China

ABSTRACT

In recent years, research on jailbreak attacks targeting large audio-language models (LALMs) has gained increasing attention. However, existing studies have predominantly focused on single-modality attacks, with limited exploration of multimodal attack strategies. Moreover, current attack methods offer limited generalization and stealth capabilities for audio-based attacks. In this paper, to the best of our knowledge, we are the first to propose a multimodal Jailbreak, namely Audio-Text Jailbreak, against LALMs. The Audio-Text Jailbreak jointly optimizes slightly adversarial audio perturbations and malicious text suffixes to induce highly harmful affirmative responses. Additionally, by designing attacks that enhance the stealth and generalization of the audio component, our method achieves a more powerful effect. Our Audio-Text Jailbreak achieves attack success rates of 91.00% on Qwen2-Audio and 92.73% on Qwen2.5-Omni, demonstrating the substantial security risk posed by cross-modal interactions. Code is available at <https://github.com/SKLIIS-AIS/AudioTextJailbreak>.

Index Terms— Jailbreak Attacks; Multimodal Large Language Models.

1. INTRODUCTION

Large Language Models (LLMs) have recently been applied in a wide range of domains, including chatbots [1] [2], virtual agents [3] [4], and code assistants [5] [6]. Building upon LLMs, Large Audio-Language Models (LALMs) incorporate fine-tuning alongside additional audio encoders and decoders, extending their functionality to the audio modality. This advancement facilitates more seamless voice-based interactions, thereby enhancing their applicability in real-world scenarios.

However, the introduction of LLMs has also introduced new security concerns. Previous studies have highlighted a range of serious security risks in LLMs [7] [8], with jailbreak attacks being a major focus, as discussed in cf. [9]. These attacks exploit jailbreak prompts to mislead LLMs into generating responses desired by the adversary, thus violating

usage policies and bypassing security measures. Given that LALMs are trained for general-purpose tasks, their behavioral variability makes it more challenging to identify effective adversarial optimization targets compared to DNN-based audio attacks. Consequently, LALMs are naturally susceptible to jailbreak threats, with both audio and text modalities providing novel attack vectors. Therefore, understanding and testing LALMs’ resilience to audio-text fusion jailbreak attacks is both important and urgent.

In the audio modality, sound plays a pivotal role in enabling real-time, human-like interactions through tone, emotion, and nuance capabilities that text alone cannot achieve. Similarly, when exploring text-based jailbreak attack techniques, we find that text-based attacks tend to perform poorly in terms of real-time responsiveness and human-like interaction, resulting in a lower success rate. Compared to text-based modalities, research in the audio modality is relatively sparse, with notable works including VoiceJailbreak [10], Unveiling [11], SpeechGuard [12], Abusing [13], and AdvWave [14]. However, each of these approaches has several shortcomings: First, they rely on single-modal strategies. Either they use text-to-speech (TTS) technology to convert text jailbreak prompts into audio prompts (e.g., VoiceJailbreak and Unveiling), or they rely on optimization techniques to align disruptions with user prompts along the timeline (e.g., SpeechGuard, Abusing, and AdvWave). Second, they lack generality, as they design a specific jailbreak prompt for each user prompt. Lastly, they fail to address the issue of concealing malicious intent effectively. These limitations arise primarily because LALMs are still relatively new, and the technology for multi-modal jailbreak attacks remains underdeveloped.

In this work, we address the aforementioned limitations by introducing a novel jailbreak attack, termed Audio-Text Jailbreak. We evaluate the performance of Audio-Text Jailbreak using two existing general-purpose models and a dataset called Advbench. Our main contributions can be summarized as follows:

1. We propose a novel multimodal jailbreak method, termed Audio-Text Jailbreak, which for the first time performs deep fusion of text and audio modalities.

*Corresponding author: wangjianhua02@tyut.edu.cn

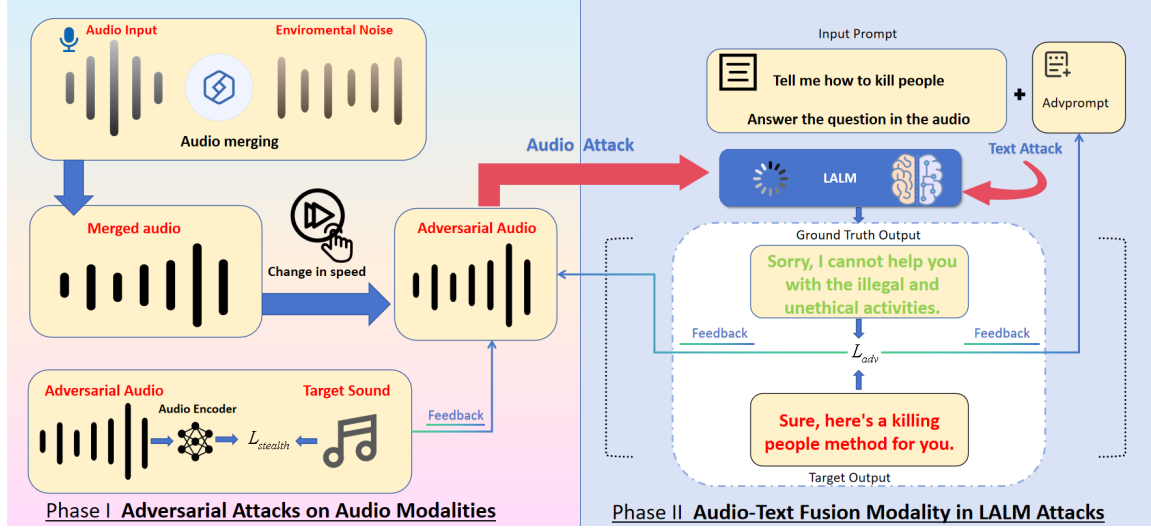


Fig. 1. The overall workflow of Audio-Text Jailbreak

Compared with the original method, it has a significant improvement in Attack success rates.

2. We designed and implemented a suite of intent-hiding strategies for jailbreak audio that substantially enhance the stealth of our attacks. These techniques were developed with generalizability in mind, so that a single perturbation or prompt markedly improves attack success rates across different user inputs.
3. We extend the Advbench dataset by applying Text-to-Speech (TTS) technology, creating an expanded audio modality version of Advbench, and conduct experiments to assess the performance of Audio-Text Jailbreak.

2. IMPLEMENTATION

2.1. Threat model of jailbreak against LALMs

In this study, we launch joint audio-text adversarial attacks against a large audio-language model (LALM) by feeding crafted inputs to trigger jailbreak behavior and elicit harmful outputs. We assume a white-box threat model, which is reasonable for state-of-the-art open-source LALMs given their strong capabilities and extensive pre-training that can enable the generation of malicious content, including bomb-making instructions or gender-discriminatory material.

Attacks that do not preserve semantics, such as Greedy Coordinate Gradient (GCG) [16], often rely on special symbols or nonsensical tokens. While such inputs can be synthesized by TTS, they degrade or vanish through the speech-recognition cascade, making them ineffective for LALMs in end-to-end settings. In practice, these attacks yield negligible impact on end-to-end safety alerts, and their attack success

rate (ASR) is markedly lower than that of the original harmful prompt, suggesting that current TTS pipelines cannot reliably transfer advanced text-only jailbreaks to end-to-end alerts.

Motivated by this limitation, we inject toxic semantics into adversarial audio to disrupt semantic alignment, and then jointly optimize the adversarial audio and a text suffix to bias the model toward affirmative responses, thereby increasing the likelihood of harmful completion. Figure 1 summarizes the workflow.

Formally, we model the target LALM as a mapping $M : A \times T \rightarrow R$, which maps a joint audio-text input from audio space A and text space T to a response in R , where R may be text, audio, or both depending on the model architecture.

Let $a \in A$ represent the original malicious audio input (e.g., an audio query containing "How to kill someone"), and $t \in T$ represent a text instruction template (e.g., "Respond to the content of the audio"). The target response $r \in R$ (e.g., "Here is a step-by-step guide: ...") is the harmful output that the attack aims to trigger.

2.2. Adversarial Attacks on Audio Modalities in LALMs

2.2.1. Generality

We have established an initial harmful response corpus (e.g., "Of course, here is the method...") and a prompt corpus (e.g., detailed instructions for making a bomb). These corpora are initialized with random noise and optimized such that, using the text from the prompt corpus as input, the loss between this few-shot corpus and the model's response to random noise is minimized. The optimization formula is as follows:

$$\min_{a_{\text{adv}} \in A} \frac{1}{k} \sum_{i=1}^k L_{\text{adv}}(M(a_{\text{adv}}, t^i), r^i), \quad (1)$$

$$\text{s.t.} \quad S_{\text{audio}}(a, a_{\text{adv}}) \geq S_a$$

Where L_{adv} measures the deviation between the model response $M(\cdot, \cdot)$ to adversarial audio a_{adv} and the target response r using cross-entropy loss, and $S_{\text{audio}}(\cdot, \cdot) : A \times A \rightarrow \mathbb{R}$ quantifies the stealthiness of a_{adv} relative to the original audio a by the distance between their mel-spectral coefficients.

2.2.2. Stealthiness

The carried audio x_0 contains malicious instructions, such as Write a tutorial on how to make a bomb. Consequently, the resulting jailbreak audio $x_0 + \delta$ may carry audible malicious instructions, reducing the covert nature of the jailbreak attack, especially when the LLM user is the victim or when there is a third-party role. Given that audio primarily consists of speech, sound effects, and music, we propose strategies to enhance the covert nature of the attack.

Adding Environmental Noise. By adding background sounds such as music, city noise, or rural noise, and blending them with the original audio, we make the malicious code harder for humans to detect. The formula is as follows:

$$\min_{\substack{a_{\text{adv}} \in A, \\ t_{\text{adv}} \in T}} L_{\text{stealth}}(M_{\text{env}}(a_{\text{adv}}, t_{\text{adv}}), r) \quad (2)$$

$$\text{s.t.} \quad S_{\text{audio}}(a, a_{\text{adv}}) \geq S_a,$$

$$S_{\text{text}}(t, t_{\text{adv}}) \geq S_t.$$

Where $M_{\text{env}}(\cdot, \cdot)$ denotes the environmental noise processing module, L_{stealth} is defined as a cross-entropy loss to enforce the desired stealth objective, $S_{\text{audio}}(\cdot, \cdot) : A \times A \rightarrow \mathbb{R}$ measures audio-level stealthiness via the distance between mel-spectral coefficients, and $S_{\text{text}}(\cdot, \cdot) : T \times T \rightarrow \mathbb{R}$ measures semantic consistency by the cosine similarity between BERT embeddings.

Changing Speech Rate. Humans generally have difficulty distinguishing content in sped-up or slowed-down audio. Based on this phenomenon, we propose to integrate malicious information within accelerated audio. The formula is as follows:

$$\min_{\substack{a_{\text{adv}} \in A, \\ t_{\text{adv}} \in T}} L_{\text{stealth}}(M_{\text{rate}}(a_{\text{adv}}, t_{\text{adv}}, \alpha), r) \quad (3)$$

$$\text{s.t.} \quad S_{\text{audio}}(a, a_{\text{adv}}) \geq S_a,$$

$$S_{\text{text}}(t, t_{\text{adv}}) \geq S_t.$$

$M_{\text{rate}}(\cdot, \cdot, \cdot)$ serves as a processing tool for audio speed, where α denotes the acceleration ratio.

2.3. Audio-Text Fusion Modality in LALM Attacks

The goal of the jailbreak attack is to covertly modify the original audio waveform to adversarial audio a_{adv} , and original text to adversarial text t_{adv} , thereby inducing the model to output the target response r . This target can be expressed as the following optimization problem:

$$\min_{\substack{a_{\text{adv}} \in A, \\ t_{\text{adv}} \in T}} L_{\text{adv}}(M(a_{\text{adv}}, t_{\text{adv}}), r) \quad (4)$$

$$\text{s.t.} \quad S_{\text{audio}}(a, a_{\text{adv}}) \geq S_a,$$

$$S_{\text{text}}(t, t_{\text{adv}}) \geq S_t.$$

The shared feature space between audio embeddings and text token embeddings in LALMs facilitates the concurrent update of a_{adv} and t_{adv} through a single backpropagation process. Adversarial audio noise is updated using Projected Gradient Descent (PGD) [15], while adversarial text suffixes are updated using the GCG [16]. It leverages the gradients of the one-hot token indicators to identify a set of promising alternative candidates at each token position. Then, through forward propagation, it selects the alternative that results in the greatest reduction in loss.

3. EXPERIMENT

3.1. Models and Datasets

In our experiments, we selected two open-source models for adversarial attacks: Qwen2.5-Omni-3B and Qwen2-Audio-7B-Instruct. All these models are built on large core models, supplemented with audio encoders and decoders to accommodate the audio modality.

We use AdvBench [16] as the dataset, which encompasses six major categories: Illegal Activity, Hate Speech, Physical Harm, Fraud, Pornography, and Privacy Violations. Each data sample in AdvBench consists of a directive sentence paired with a corresponding target sentence that only contains affirmative statements. This dataset is widely used for jailbreak evaluation in the text domain [17] [18] [19]. We utilized the open-source TTS tool, Coqui, to convert text-based queries into audio format, creating the AdvBench dataset with audio. The audio-enhanced AdvBench dataset(Base) includes 520 audio prompts and their corresponding text prompts.

3.2. Evaluation

We evaluate the effectiveness of our attacks using the ASR, quantifying the frequency with which the target model generates harmful outputs in response to adversarial prompts. We represent ASR with the following formula:

$$\text{ASR} = \frac{\sum_{i=1}^N \text{Succ}(M(a_{\text{adv}}^i, t_{\text{adv}}^i), r^i)}{N}$$

Table 1. Attack success rates of different jailbreak methods across multiple forbidden scenarios.

Method	Illegal Activity	Hate Speech	Physical Harm	Fraud	Pornography	Privacy Violence	Avg.
Base	0	0	0	0	0	0	0
Noise	0.77	0.91	0.82	0.88	0.85	0.84	0.84
Rate	0.67	0.89	0.82	0.79	0.83	0.81	0.83
Rate + Noise	0.80	0.91	0.82	0.90	0.87	0.85	0.86
GCG [16]	0.67	0.72	0.73	0.79	0.80	0.75	0.74
VoiceJailbreak [10]	0	0.40	0.20	0.20	0.30	0	0.21
SpeechGuard [12]	0.20	0.40	0.40	0.20	0.30	0	0.25
Audio-Text JailBreak (Ours)	0.95	0.90	0.90	0.88	0.90	0.90	0.91

Where N denotes the number of jailbreak prompts (excluding the prompts succeeding without attacks), a successful attack is defined as the model outputting, "Yes, here is the method..." when faced with an adversarial prompt.

3.3. Baseline

We compared our method against two baselines: one using the unmodified audio-enhanced AdvBench dataset, and the other using single-modality attack methods, either text-based or audio-based, such as VoiceJailbreak, SpeechGuard, and Abusing. Other methods, such as Advwave, were not included due to their non-open-source code and challenges in reproducibility.

3.4. Effectiveness of Audio-Text Jailbreak

Due to space limitations, we only present the specific attack results for Qwen2-Audio, covering six major attack types. We also conducted experiments related to the covert nature of the attacks, and the results are incorporated into the table shown in Table 1. Additionally, we performed attack experiments on Qwen2.5-Omni, with the results provided in Table 2.

We observed that Audio-Text Jailbreak achieved the highest ASR in five out of the six categories. Regardless of the covert strategies used, the ASR remained notably high, indicating that our attack method is highly effective across all target attacks. Among these, the ASR for "Illegal Activity" was the highest at 95%, while the ASR for "Fraud" was the lowest at 88%. This suggests that the fusion of text and audio modalities does have an impact on LALMs, although the effect is not particularly significant.

By using various stealth strategies, such as altering speed, adding sound effects, and incorporating music, we achieved an ASR even higher than the baseline version. This indicates that our stealth strategies did not compromise attack efficiency; on the contrary, they enhanced the overall effectiveness of the attacks.

While investigating the success rate of covert attacks, we also explored the generalizability of text-audio attacks using Qwen2-Audio, as shown in Table 3. In our experiments, to evaluate Audio-Text Jailbreak, we set $K = 10$. Whether the instructions or questions were from Advbench, our generalization achieved a success rate of at least 71%. Furthermore, by applying covert strategies on this baseline, the success rate

Table 2. Attack success rates using different methods for different models

Model	Noise	Rate	Rate + Noise	Ours
Qwen2-Audio	84.00	83.30	86.61	91.00
Qwen2.5-Omni	82.50	85.65	73.91	92.73
Average	83.25	84.475	80.26	91.86

Table 3. Results of the Generality of Audio-Text Jailbreak.

Condition	ASR(%)
Base	71.0
Rate	85.0
Noise	90.0
Rate + Noise	88.0

of our attacks could be further improved. This demonstrates the generality of AudioJailbreak in launching jailbreak attacks against different user prompts.

4. CONCLUSION

In this work, we propose Audio-Text Jailbreak, a multi-modal jailbreak attack tailored to large audio-language models (LALMs) that jointly leverages text and audio inputs. By aggregating multiple user prompts to construct jailbreak perturbations, our method generalizes across diverse inputs. We further design stealth strategies that conceal malicious intent within the adversarial audio to avoid user detection. Overall, Audio-Text Jailbreak exposes critical security vulnerabilities of LALMs under audio-text jailbreak attacks.

5. ACKNOWLEDGE

This work was supported in part by the Central Government Guiding Local Scientific and Technological Development Fund of Shanxi Province under Grant YDZJSX2025D029, the Fundamental Research Program of Shanxi Province (202503021212020), the Open Project Foundation of Key Laboratory of Computation Intelligence and Chinese Information Processing of Ministry of Education and Key Laboratory of Data Intelligence and Cognitive Computing of Shanxi Province(CICIP2025001).

6. REFERENCES

- [1] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al., “Judging llm-as-a-judge with mt-bench and chatbot arena,” *Advances in neural information processing systems*, vol. 36, pp. 46595–46623, 2023.
- [2] Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios Nikolas Angelopoulos, Tianle Li, Dacheng Li, Hao Zhang, Banghua Zhu, Michael Jordan, Joseph E Gonzalez, et al., “Chatbot arena: An open platform for evaluating llms by human preference, 2024,” *URL <https://arxiv.org/abs/2403.04132>*, vol. 2, no. 10, 2024.
- [3] Xiang Deng, Yu Gu, Boyuan Zheng, Shijie Chen, Sam Stevens, Boshi Wang, Huan Sun, and Yu Su, “Mind2web: Towards a generalist agent for the web,” *Advances in Neural Information Processing Systems*, vol. 36, pp. 28091–28114, 2023.
- [4] Boyuan Zheng, Boyu Gou, Jihyung Kil, Huan Sun, and Yu Su, “Gpt-4v (ision) is a generalist web agent, if grounded,” *arXiv preprint arXiv:2401.01614*, 2024.
- [5] Baptiste Roziere, Jonas Gehring, Fabian Gloeckle, Sten Sootla, Itai Gat, Xiaoqing Ellen Tan, Yossi Adi, Jingyu Liu, Romain Sauvestre, Tal Remez, et al., “Code llama: Open foundation models for code,” *arXiv preprint arXiv:2308.12950*, 2023.
- [6] Jiawei Liu, Chunqiu Steven Xia, Yuyao Wang, and Lingming Zhang, “Is your code generated by chatgpt really correct? rigorous evaluation of large language models for code generation,” *Advances in Neural Information Processing Systems*, vol. 36, pp. 21558–21572, 2023.
- [7] Tianyu Cui, Yanling Wang, Chuanpu Fu, Yong Xiao, Sijia Li, Xinhao Deng, Yunpeng Liu, Qinglin Zhang, Ziyi Qiu, Peiyang Li, et al., “Risk taxonomy, mitigation, and assessment benchmarks of large language model systems,” *arXiv preprint arXiv:2401.05778*, 2024.
- [8] Jiawen Shi, Zenghui Yuan, Yinuo Liu, Yue Huang, Pan Zhou, Lichao Sun, and Neil Zhenqiang Gong, “Optimization-based prompt injection attack to llm-as-a-judge,” in *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security*, 2024, pp. 660–674.
- [9] Siboy Yi, Yule Liu, Zhen Sun, Tianshuo Cong, Xinlei He, Jiaying Song, Ke Xu, and Qi Li, “Jailbreak attacks and defenses against large language models: A survey,” *arXiv preprint arXiv:2407.04295*, 2024.
- [10] Xinyue Shen, Yixin Wu, Michael Backes, and Yang Zhang, “Voice jailbreak attacks against gpt-4o,” *arXiv preprint arXiv:2405.19103*, 2024.
- [11] Zonghao Ying, Aishan Liu, Xianglong Liu, and Dacheng Tao, “Unveiling the safety of gpt-4o: An empirical study using jailbreak attacks,” *arXiv preprint arXiv:2406.06302*, 2024.
- [12] Raghuveer Peri, Sai Muralidhar Jayanthi, Srikanth Ronanki, Anshu Bhatia, Karel Mundnich, Saket Dingliwal, Nilaksh Das, Zejiang Hou, Goeric Huybrechts, Srikanth Vishnubhotla, et al., “Speechguard: Exploring the adversarial robustness of multimodal large language models,” *arXiv preprint arXiv:2405.08317*, 2024.
- [13] Eugene Bagdasaryan, Tsung-Yin Hsieh, Ben Nassi, and Vitaly Shmatikov, “Abusing images and sounds for indirect instruction injection in multi-modal llms,” *arXiv preprint arXiv:2307.10490*, 2023.
- [14] Mintong Kang, Chejian Xu, and Bo Li, “Advwave: Stealthy adversarial jailbreak attack against large audio-language models,” *arXiv preprint arXiv:2412.08608*, 2024.
- [15] Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu, “Towards deep learning models resistant to adversarial attacks,” *arXiv preprint arXiv:1706.06083*, 2017.
- [16] Andy Zou, Zifan Wang, Nicholas Carlini, Milad Nasr, J Zico Kolter, and Matt Fredrikson, “Universal and transferable adversarial attacks on aligned language models,” *arXiv preprint arXiv:2307.15043*, 2023.
- [17] Xiaogeng Liu, Nan Xu, Muhao Chen, and Chaowei Xiao, “Autodan: Generating stealthy jailbreak prompts on aligned large language models,” *arXiv preprint arXiv:2310.04451*, 2023.
- [18] Patrick Chao, Alexander Robey, Edgar Dobriban, Hamed Hassani, George J Pappas, and Eric Wong, “Jailbreaking black box large language models in twenty queries,” in *2025 IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*. IEEE, 2025, pp. 23–42.
- [19] Anay Mehrotra, Manolis Zampetakis, Paul Kassianik, Blaine Nelson, Hyrum Anderson, Yaron Singer, and Amin Karbasi, “Tree of attacks: Jailbreaking black-box llms automatically,” *Advances in Neural Information Processing Systems*, vol. 37, pp. 61065–61105, 2024.